

4.1

$$\begin{aligned} \text{a. } SST &= \sum (y_i - \bar{y})^2 = \sum y_i^2 - N\bar{y}^2 \\ &= 7825.94 - 20 \times 19.21^2 = 445.458 \end{aligned}$$

$$R^2 = SSR/SST = \frac{375.47}{445.458} \approx 0.843$$

$$\text{b. } R^2 = 1 - \frac{SSE}{SST} \Rightarrow SSE = (1 - R^2)SST \approx 151.65$$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{151.65}{18} \approx 8.425$$

$$\text{c. } \sum \hat{e}_i^2 = SSE$$

$$\Rightarrow R^2 = 1 - \frac{182.85}{631.63} \approx 0.7105$$

4.3

$$\text{a. } \hat{y}_i = 1.2 + 0.8x_i$$

$$x_0 = 4 \text{ 代 } \lambda$$

$$\hat{y}_0 = 1.2 + 3.2 = 4.4$$

$$\text{b. } se(y) = \sigma \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

$$\bar{x} = 1$$

$$\sigma = \sqrt{\frac{SSE}{N-2}} = \sqrt{1.2}$$

$$se(y) = \sqrt{1.2} \sqrt{\frac{1}{5} + \frac{(4-1)^2}{10}} \approx 1.1489$$

$$\text{c. } se(f) = \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

$$= \sqrt{1.2} \sqrt{1 + \frac{1}{5} + \frac{(4-1)^2}{10}} \approx 1.5875$$

$$\begin{aligned} 95\% \text{ PI for } \hat{y}_0 &: \hat{y}_0 \pm t_{0.025, 3} \cdot \text{se}(f) \\ &= [-1.052, 9.052] \end{aligned}$$

$$\begin{aligned} \text{d. } 99\% \text{ PI for } \hat{y}_0 &: \hat{y}_0 \pm t_{0.005, 3} \cdot \text{se}(f) \\ &= [-5.272, 13.272] \end{aligned}$$

$$\text{e. } \bar{x} = 1 \Rightarrow \text{se}(f) = \sqrt{1.2} \sqrt{1 + \frac{1}{5} + \frac{(1-1)^2}{10}} = 1.2$$

95% PI for y given $x = \bar{x}$:

$$\hat{y} \pm t_{0.025, 3} \cdot \text{se}(f) = [-2.819, 4.819]$$

The width of the interval in e. is smaller than the interval in c. It is because when $x_0 = \bar{x}$, it is closet to the sample center .